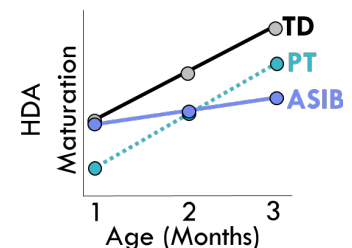


SPECIFIC AIMS

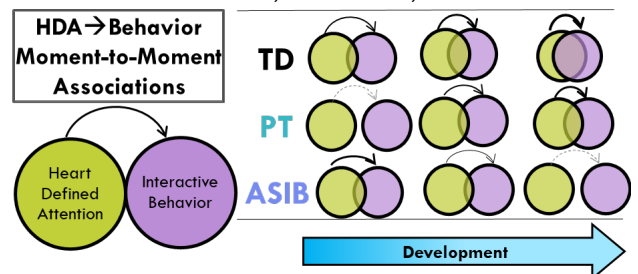
Atypical attention is among the earliest-emerging, lifelong features of autism spectrum disorder (ASD) and a particularly compelling candidate for an ASD biomarker¹⁻⁴. Our work, and that of others, suggests that disrupted attention may be detectable and developmentally predictive in early infancy⁵⁻⁸, yet the biological underpinnings of this disruption remain unknown. The autonomic nervous system (ANS) is an early-emerging bioregulatory system critical for supporting neonatal arousal and infant attention⁹⁻¹¹. Early disruptions in ANS's regulation of attention ("autonomic regulation of attention") may have cascading consequences for interaction, learning, and development^{8,9,11}. Broad ANS dysfunction is observed in older children already diagnosed with ASD¹², but there remains a tremendous gap in knowledge regarding **whether** and **when** ANS dysfunction contributes to attention abnormalities and how infant autonomic regulation of attention predicts the emergence of ASD symptoms.

The current study fills this gap by examining autonomic regulation of attention in the neonatal period and the developmental consequences therein for the emergence of ASD symptoms in infant siblings of children with ASD (**ASIBs**), infants born preterm (**PTs**), and typically developing (**TD**) infants. A critical innovation of this study is leveraging a comparison group of PTs who, intriguingly, experience global ANS dysfunction from birth^{9,13-15}. While PTs show slightly elevated rates of ASD (~7%) compared to the general population (~2%)¹⁶, the vast majority of PTs do *not* develop ASD. In this study, **inclusion of PT infants as a model for global ANS dysfunction** will help investigate our hypothesis that early and broad ANS dysfunction (not specific to attention) *does not* lead to ASD-specific symptoms. This developmental process (broad ANS dysfunction → non-ASD specific outcomes) will be contrasted to **ASIBs** who experience significantly higher rates of ASD (~20%)¹⁷. Contrasting **PTs** to **ASIBs** will help delineate how *attention-specific* ANS dysfunction predicts the emergence of *ASD-specific* symptoms. Thus, our innovative **three-group research design** will compare **ASIBs** to two control groups (**PTs** and **TD** infants) to determine the specificity of autonomic regulation of attention as a key, early emerging feature associated with ASD, with incredible translational potential for a cost-effective biomarker.

Aim 1. Compare the maturation of autonomic regulation of attention in the early infant period (1-3m) across ASIBs, PTs, and TDs. Attention in the first months of life is supported by flexible modulation of arousal and alertness, regulated by the ANS¹⁸⁻²⁰. Autonomic regulation allows for attention to salient features of the environment and inhibition of responses to irrelevant (internal or external) stimuli. ANS regulation of attention is vital for learning and can be measured using **heart-defined attention (HDA)** (i.e., coordination of heart rate changes within moments of visual attention^{21,22}). Maturation of ANS and attention systems across the first months of life lead to increasing coordination of heart rate with attention. Our pilot work with ASIBs shows differences in HDA quantity (%time in HDA) and quality (magnitude of heart rate deceleration)^{23,24}. Building upon this work, Aim 1 will test the hypothesis that **ASIBs** show **typical HDA initially** compared to **TDs**. Then from 1-3m, as ANS commands an increasingly larger role in regulating attention, the development of HDA for **ASIBs** will attenuate. In contrast, **PTs** will show **delayed HDA maturation** from 1-3m (adjusted age) due to broad ANS dysfunction²⁵.



Aim 2. Determine the moment-to-moment influence of autonomic regulation of attention on interactive behavior across the first year (6m, 9m, 12m). Autonomic regulation of attention, measured via heart activity and contextualized within moment-by-moment behavior, has extraordinary potential to transform early identification of ASD by revealing specific mechanistic relations between ANS, attention, and interactive behavior. Our preliminary data suggests: (1) temporal coordination between HDA and interactive behavior in **TDs** and (2) reduced HDA-behavior coordination in **ASIBs**. Aim 2 will test the hypothesis that **ASIBs** show weakening predictive associations between HDA and interactive behavior from 6-12 months, as early symptoms of ASD emerge²⁶. In contrast, we expect **PTs** to show initially weak associations that strengthen over time (with ANS and behavioral maturation)²⁵.



Aim 3. Predict ASD symptoms at age 3 years from infant autonomic and attentional features.

ASD biomarker discovery requires quantitative modeling techniques that adequately capture the complexity of biological processes involved in such a multifaceted, heterogeneous neurodevelopmental disorder^{27,28}. Aim 3 will leverage machine learning techniques to develop a biomarker predicting the later emergence of ASD symptoms based on early (1-12 months) features of the autonomic and attentional systems (variables from Aims 1-2). We hypothesize that features related to infant autonomic regulation of attention will predict ASD symptoms.